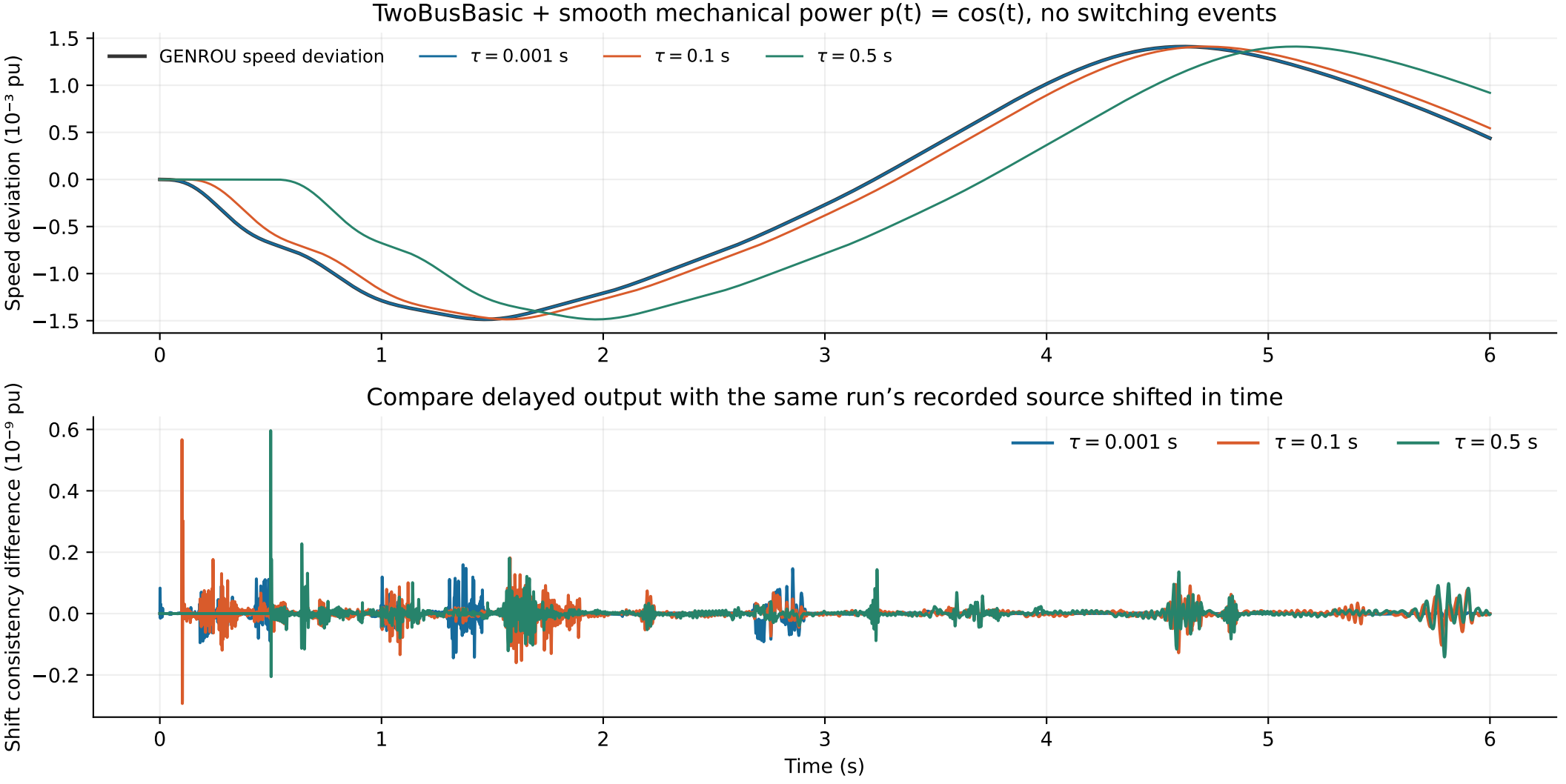
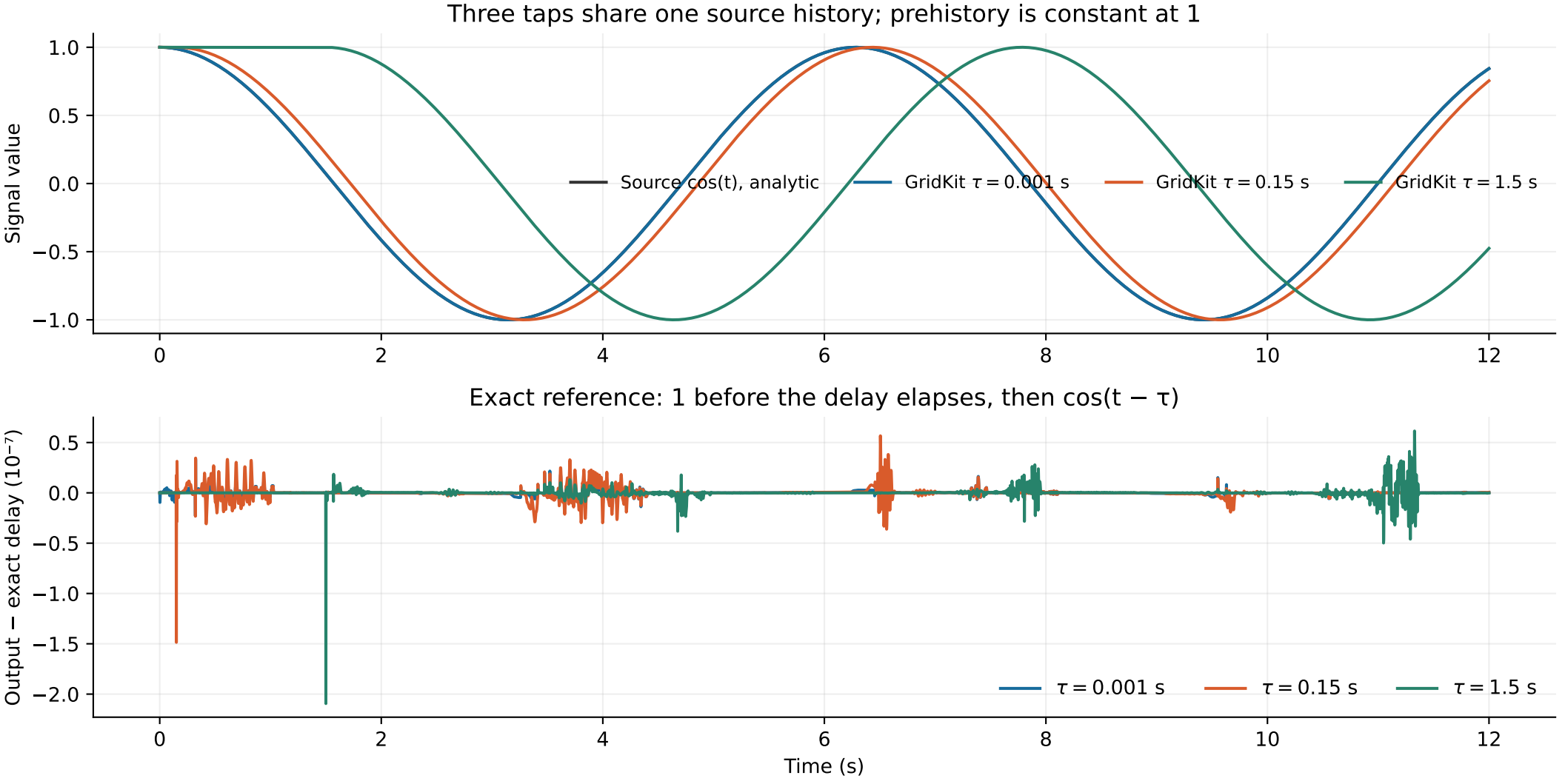
01 A generator signal delayed inside an actual grid case



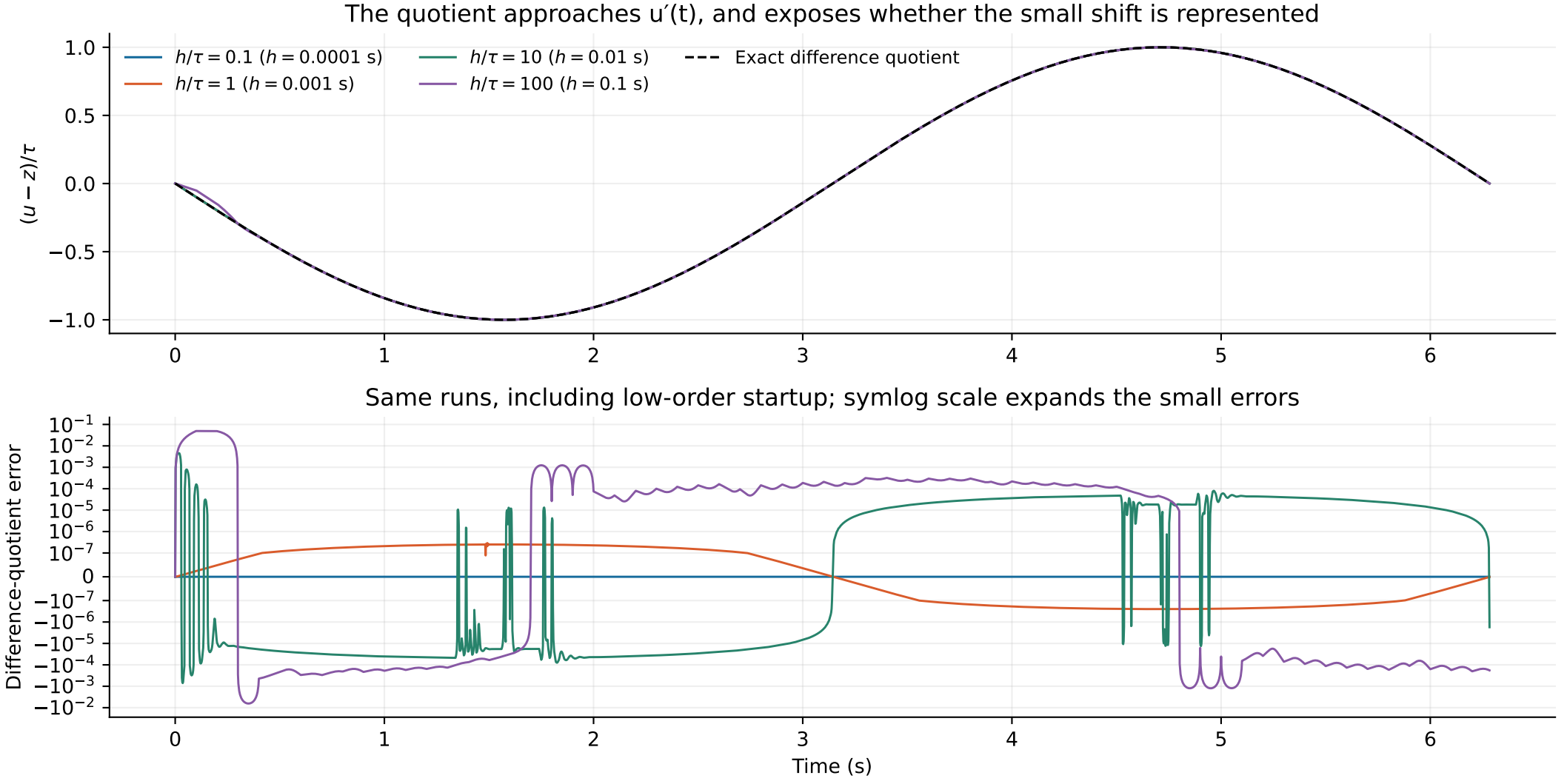
Actual DynamicSimulation CSV. Relative tolerance 10^{-8}; monitor interval 0.5 ms.

02 The JSON delay declarations, with an analytic reference



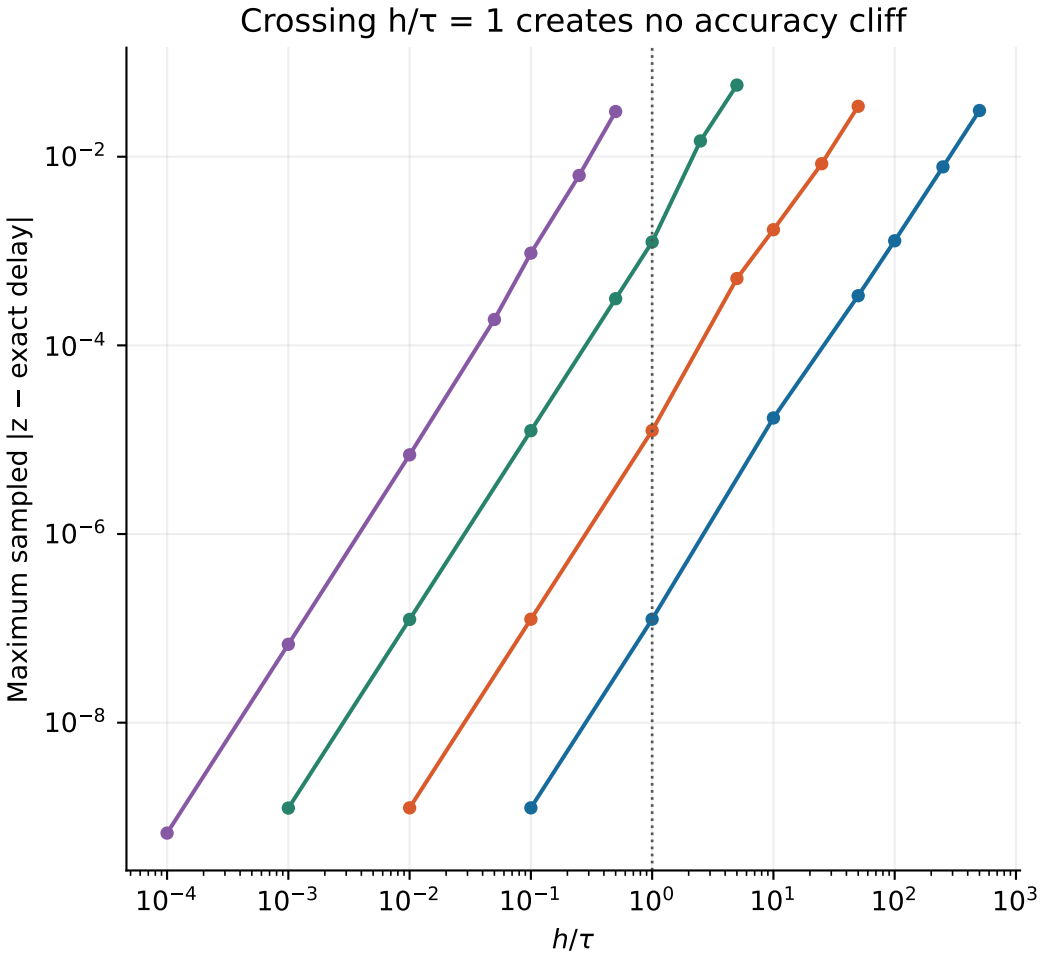
Actual DynamicSimulation CSV. Relative tolerance 10^{-7}; monitor interval 2 ms.

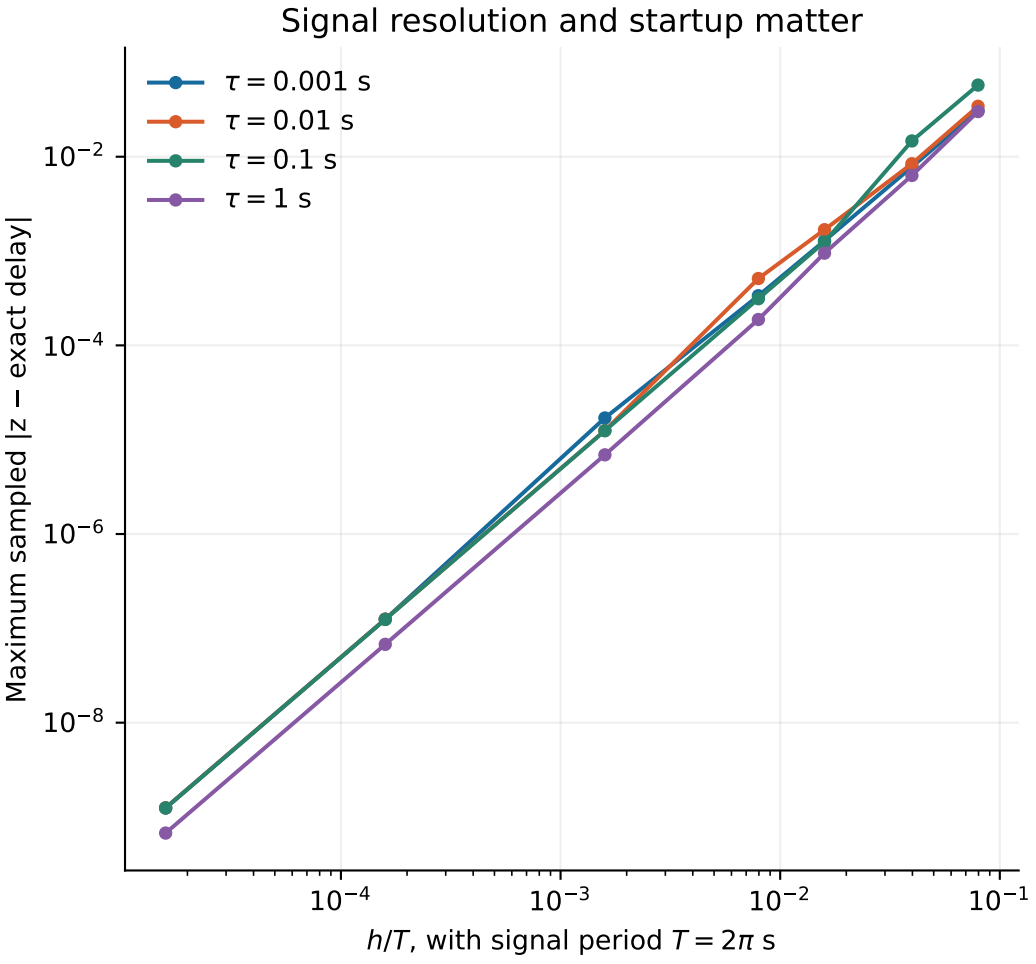
03 Make a 1 ms delay visible: inspect the difference from its source



Fixed-step diagnostic sweep; loose history tolerance allows the selected steps.

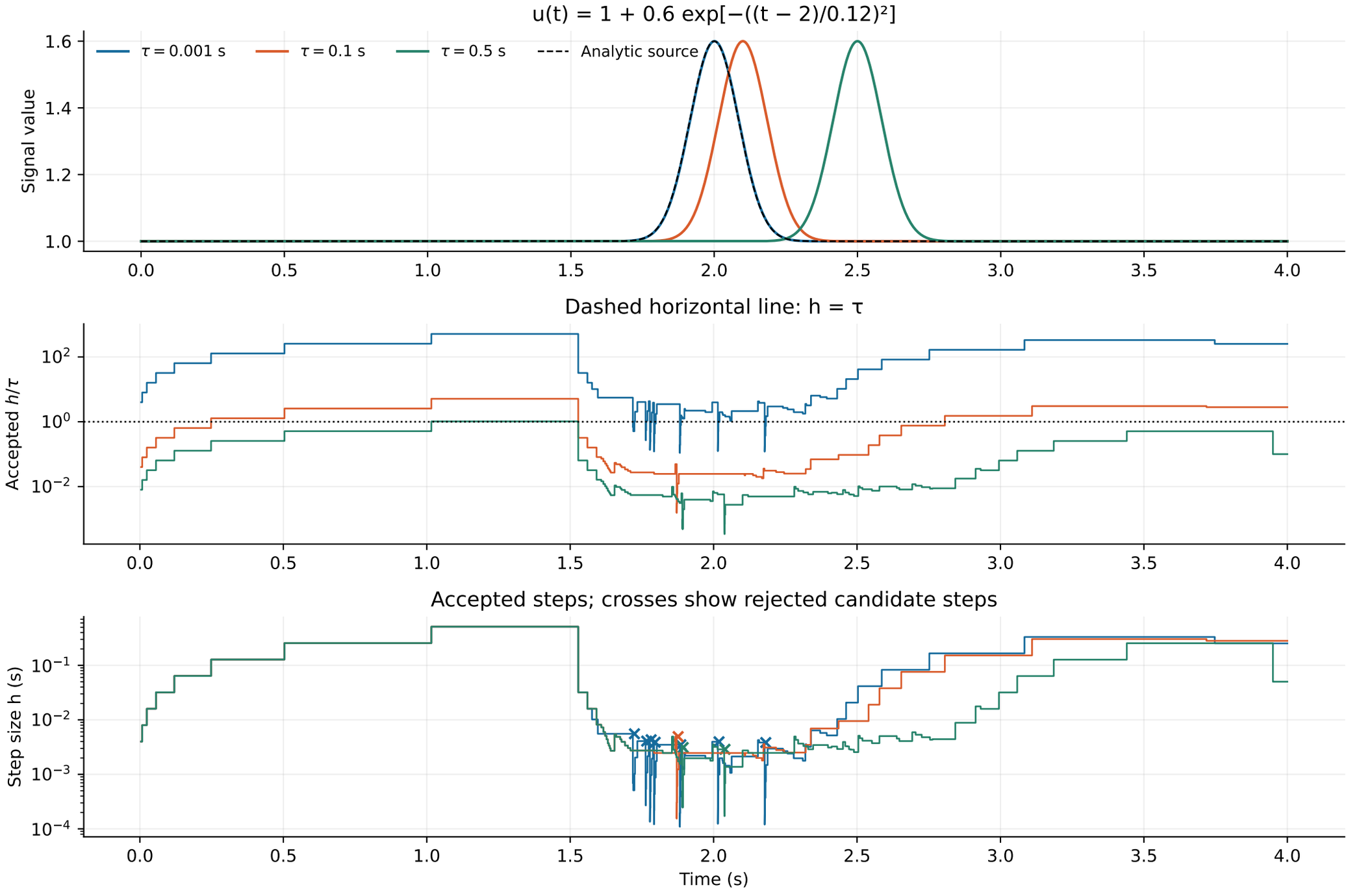
04 Step / delay ratio does not determine accuracy by itself





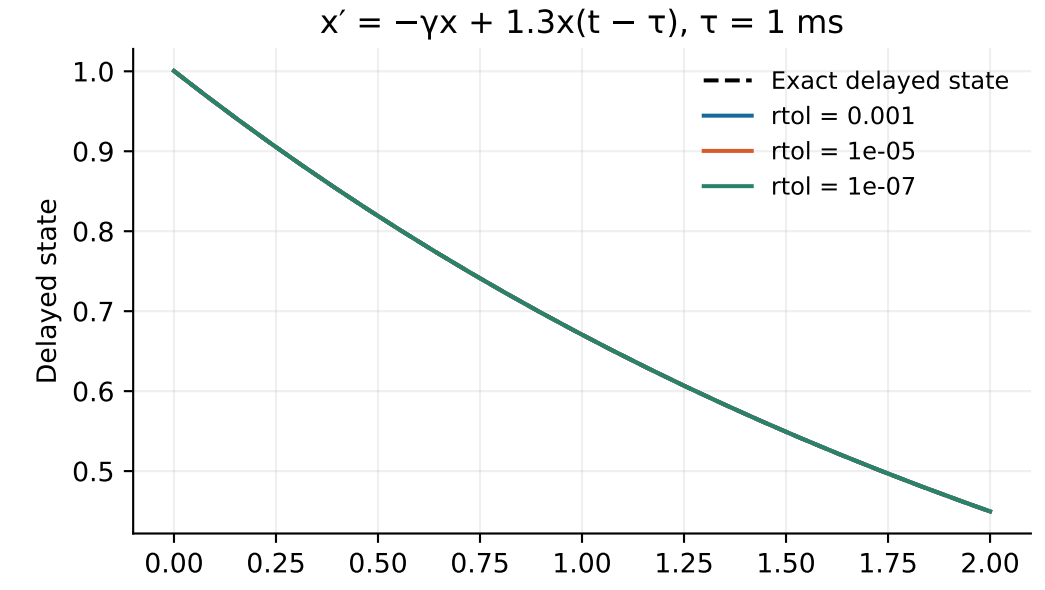
Error includes uniform monitor samples AND the midpoint of every accepted step.

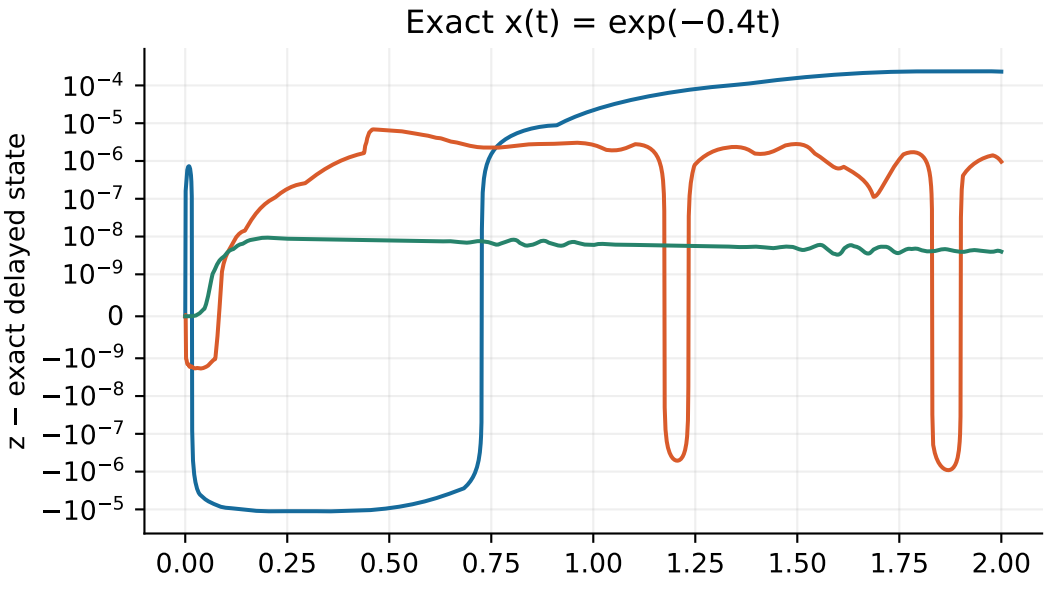
05 Adaptive steps follow a smooth pulse, without a shortest-delay cap

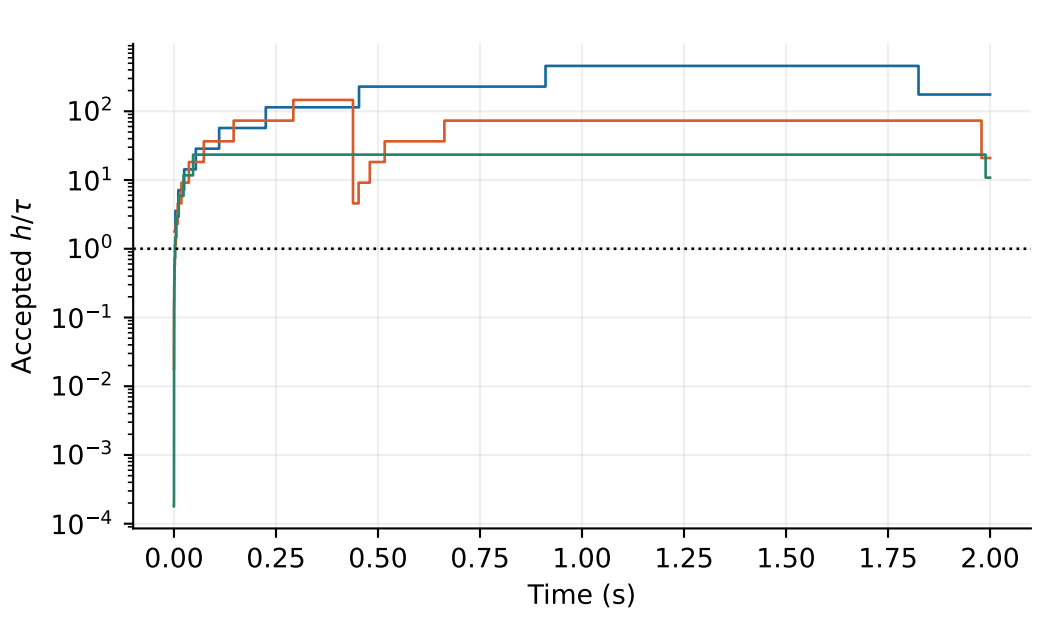


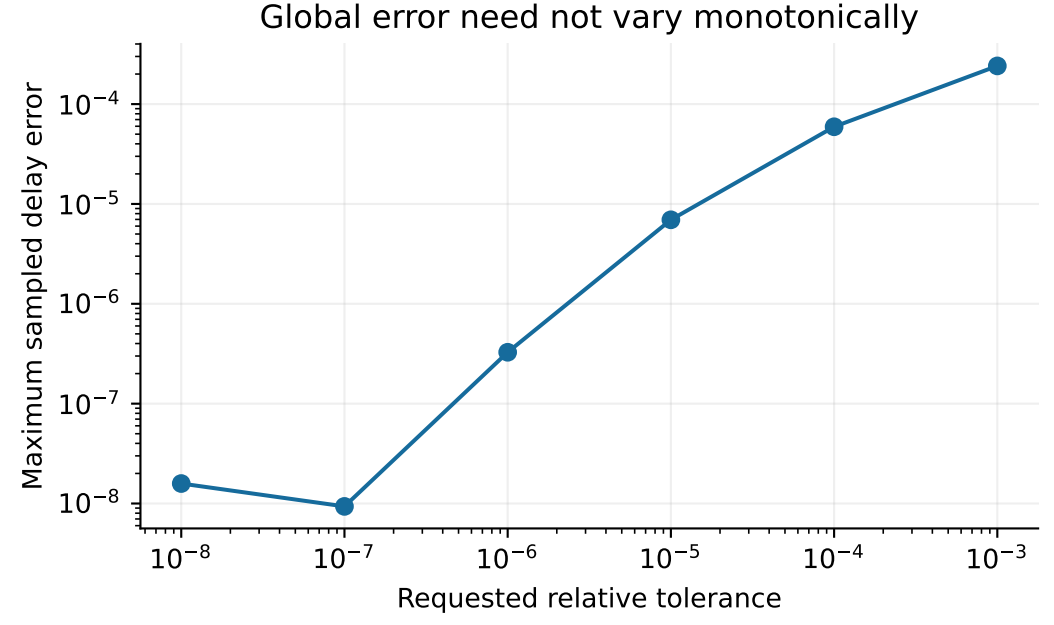
Relative tolerance 10^{-6}; absolute tolerance 10^{-8}; analytic prehistory; compression off.

06 Delay inside feedback: check against an independent exact solution



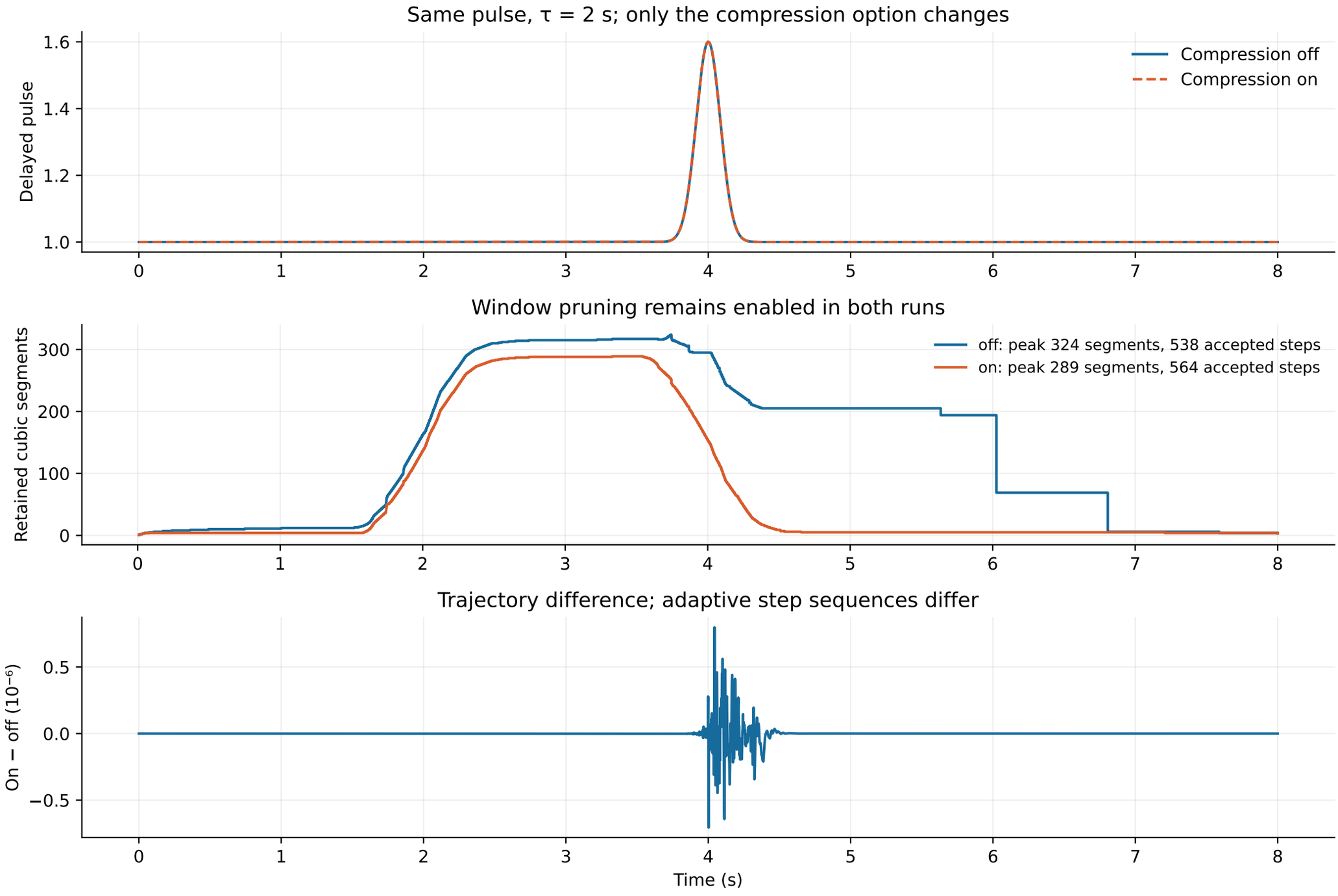






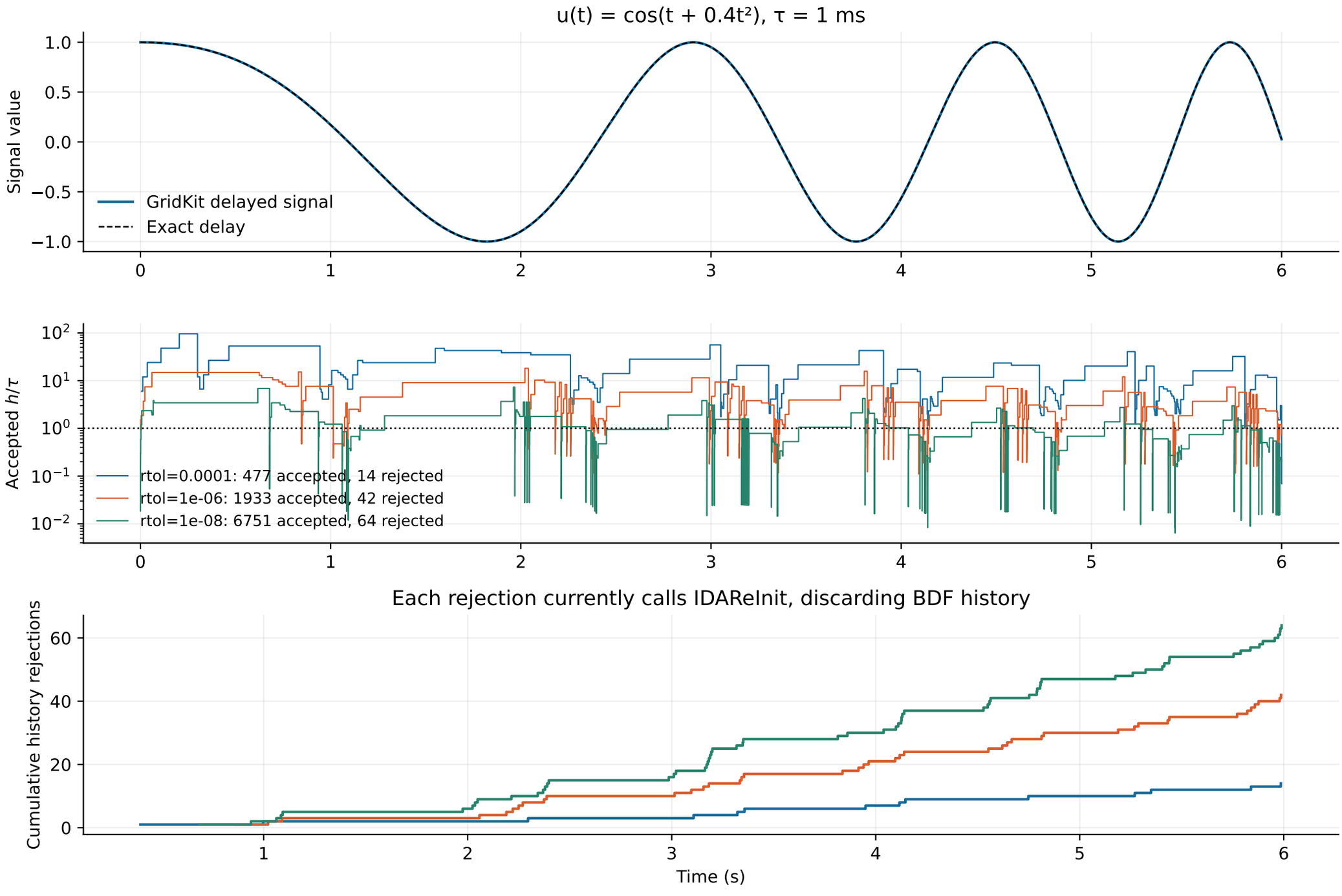
γ = 1.3 exp(0.4τ) + 0.4; exact exponential prehistory; compression off.

07 Does history compression pay for itself here?



Peak segments: 324 → 289 (10.8% reduction). Accepted steps: 538 → 564.

08 A smooth chirp exposes the cost of history-triggered retries



Smooth analytic prehistory; compression off; matched solver/history tolerances.